

Title:
Cyclistic Bike-Share: Strategic Growth & Conversion Analysis
(2025-26)

Analyzing 5.68 Million Trips to Drive Membership Conversion

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Phase 1: The Business Task (ASK)

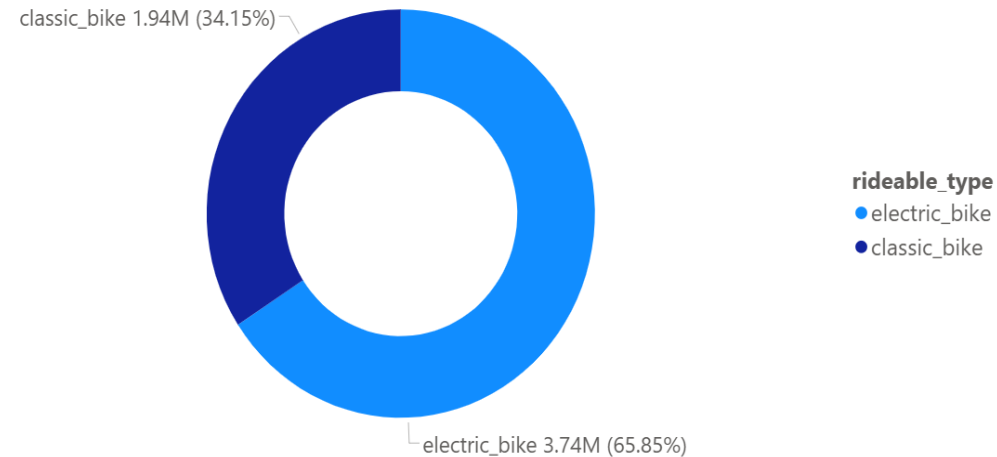
- Primary Objective:** To analyze historical trip data to identify behavioral trends and differences between "Casual" riders and "Annual Members."
 - Business Goal:** Design marketing strategies aimed at converting high-volume casual riders into loyal annual members to ensure long-term growth.
 - The Core Question:** How do annual members and casual riders use Cyclistic bikes differently?
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- Stakeholders:**
 - Lily Moreno:** Director of Marketing (Lead).
 - Cyclistic Executive Team:** Decision-makers for the marketing program.
 - Marketing Analytics Team:** Responsible for data collection and analysis.

Phase 2 & 3: Data Preparation & Processing

- ▶ **Content Points:**
- ▶ **Data Source:** 12 months of historical trip data (June 2025 – May 2026).
- ▶ **Scale:** Analyzed over **5.68 Million** records using SQL (PostgreSQL).
- ▶ **Cleaning Process:**
 - ▶ Removed duplicate *ride_id* entries for data integrity.
 - ▶ Handled NULL values in station names and coordinates.
 - ▶ Filtered out trips less than 1 minute and longer than 24 hours (outliers).
- ▶ **Feature Engineering:** Calculated *ride_length_minutes* and identified *day_of_week* to uncover behavioral patterns.

Phase 4 & 5: Analysis & Key Findings

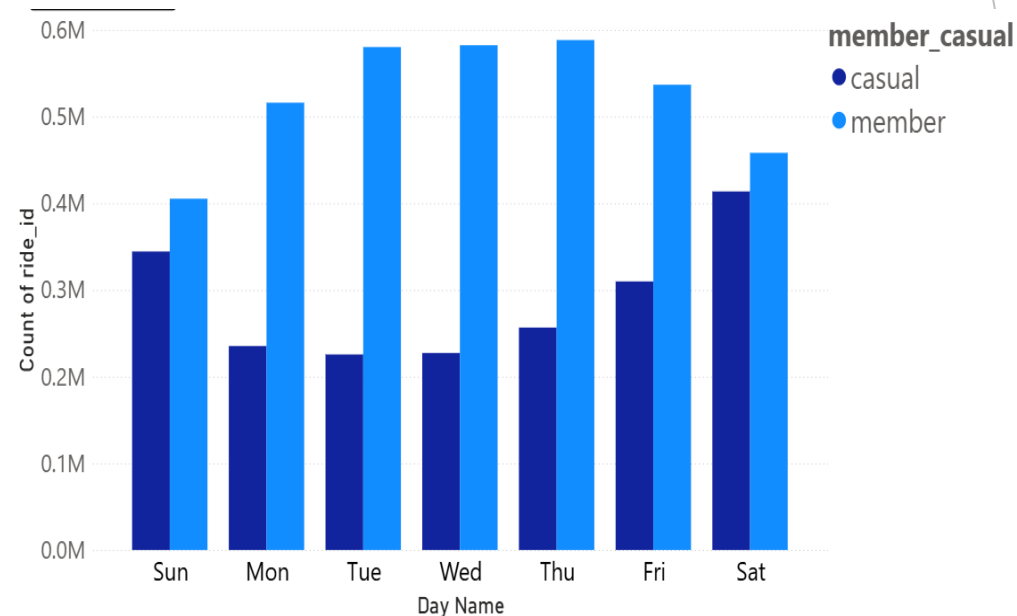
- **Total Rides Analyzed:** 5,681,150
- **Membership Breakdown:**
- **Annual Members:** 3,667,949 rides (**64.5%**)
- **Casual Riders:** 2,013,201 rides (**35.5%**)
- **Key Insight:** While Annual Members dominate the total ridership, Casual Riders still represent a massive portion (**over 2 million rides**) of the business, indicating a significant conversion opportunity.



"Cyclistic: Annual Members vs. Casual Riders"

Ride Patterns: Weekly Usage Trends

- **Casual Rider Trend:** Casual riders show significantly higher usage during weekends, peaking on Saturdays.
- **Member Rider Trend:** Annual members maintain a consistent and stable ridership throughout the weekdays.
- **Key Insight:** The weekend-focused trend suggests that casual riders primarily use Cyclistic bikes for leisure and recreational purposes.



"Casual vs. Member: Weekly Ride Patterns"

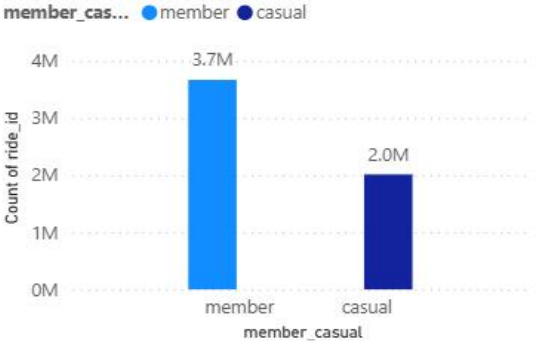
Cyclist Ride Share Analysis 2025-26

Members vs Casual

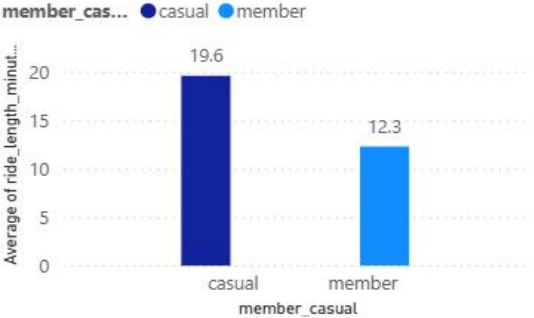
casual

member

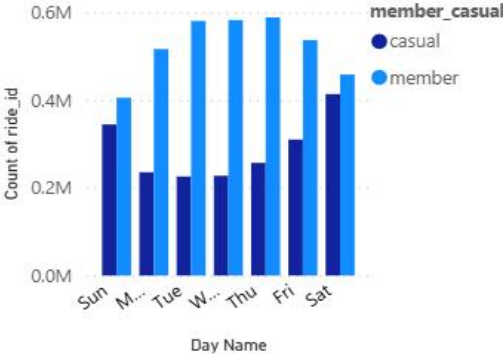
Members vs Casual Riders



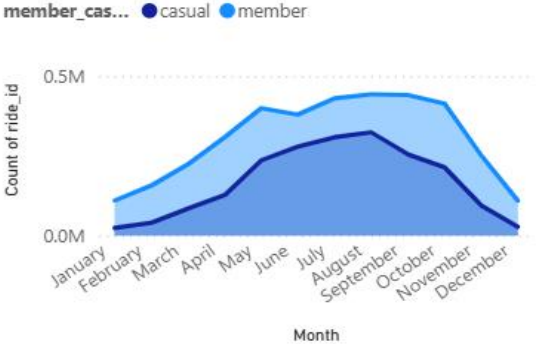
Average Ride Duration (Minutes) by Member Type



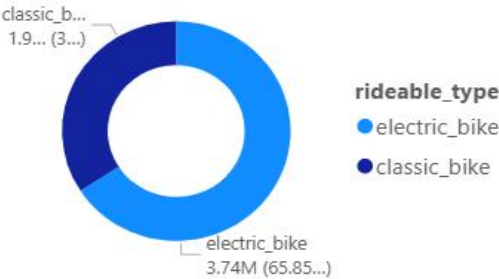
Total Numbers of Riders by Day of the week



Monthly Rides Trend: Member vs Casual



Distribution of Rides by Bike Type



Map of Popular Ride Start Locations by Member Type



Data-Driven Insights & Strategic Recommendations

▶ Key Insights (Based on Dashboard Analysis)

- ▶ **Ride Volume & Membership:** Annual members hold the majority share with **3.7M rides**, compared to **2.0M rides** by casual riders.
- ▶ **Ride Duration:** Casual riders spend significantly longer on each trip (**20 minutes** on average) compared to annual members (**12 minutes**), suggesting recreational use.
- ▶ **Weekly Trends:** Casual ride activity peaks during weekends, whereas annual members demonstrate consistent usage throughout the work week.
- ▶ **Monthly Seasonality:** Both rider groups show a surge in usage during the summer months (June–August) with a notable decline during winter.
- ▶ **Geographic & Bike Preference:** Casual riders frequent tourist and leisure-oriented zones, utilizing both electric and classic bikes for their trips.

▶ Strategic Recommendations

- ▶ **Seasonal Marketing:** Launch targeted digital ad campaigns between June and August to convert casual summer riders into annual members.
- ▶ **Incentivized Conversion:** Introduce a "Weekend-to-Weekday" trial program that offers discounted weekday access to casual users, encouraging them to adopt the service for daily commuting.
- ▶ **Location-Specific Outreach:** Place QR codes and membership information at high-traffic leisure hubs identified in the map analysis to capture casual users where they ride most.
- ▶ **Value-Based Membership:** Promote the cost-efficiency of annual memberships by highlighting the total annual savings for frequent commuters, directly contrasting their current usage patterns.

“THANK YOU”

"Through data-driven insights, we have identified that converting casual riders into annual members is the key to Cyclistic's sustainable growth. By targeting summer weekend peaks and leisure hubs, we can drive maximum conversion."

BY:

TOOBA ARSHAD

Google Data Analytics Capstone Project